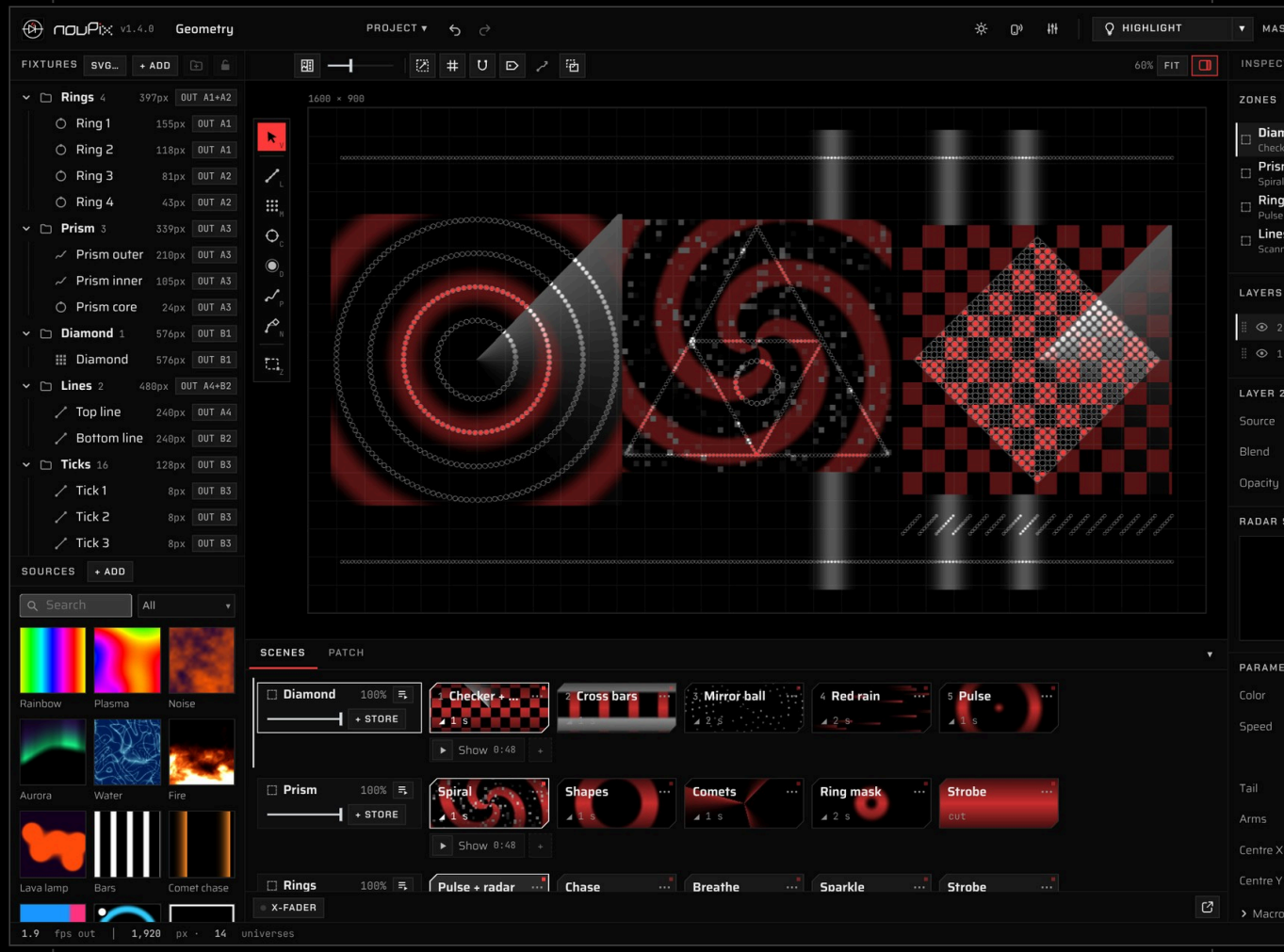
nouPix

GETTING STARTED · V1.4.0

FROM ZERO TO LIGHT, STEP BY STEP.

Install nouPix, draw your fixtures, patch them, play content in zones and layers, store scenes, and run the show from a controller or a phone.

nouPix 1.4.0 · macOS · Windows · Linux · Raspberry Pi
Free and open source · GNU AGPL-3.0 · NOULED.STUDIO
noupix.nouled.com



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START HERE Steps 03 to 17 build one small project from an empty canvas: four bars, a matrix, a ring and a wave, in two zones. Every picture is the real app.

01

INSTALL AND OPEN

GET NOUPIX RUNNING

MACOS

macOS 12 or newer · Apple silicon or Intel

1. Download the **.dmg** for your Mac from the download page.
2. Open it and drag **nouPix** onto Applications.
3. First start: click **Done** on the warning, then **System Settings → Privacy & Security → Open Anyway**.

WINDOWS

Windows 10 or newer · 64-bit

1. Download the **installer .exe**.
2. If SmartScreen stops it: **More info → Run anyway**.
3. Open nouPix from the Start menu. Allow it through the firewall on **private networks**, or node discovery won't find your controllers.

LINUX · PI

Any computer · Python 3.8+ and Chrome or Chromium

1. Get the source and run it:

```
$ git clone https://github.com/nouled-studio/nouPix.git
$ cd nouPix
$ python3 noupix.py
```

2. nouPix opens in its own window. Keep the terminal open: closing it stops the output.

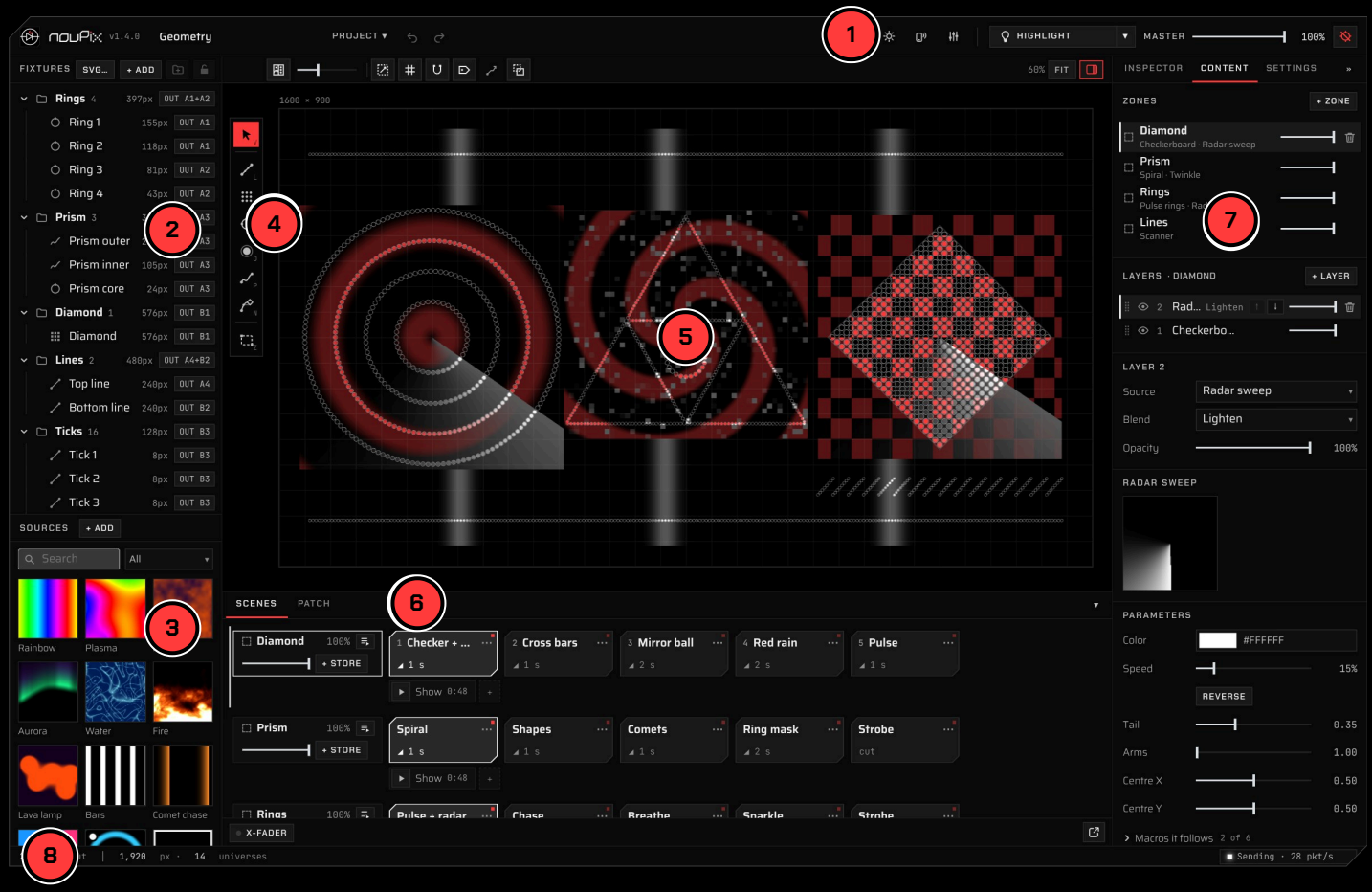
FIRST START nouPix opens with a demo layout, already playing. Look around, then start your own project (step 03).

02

THE WINDOW

EVERYTHING ON ONE SCREEN

1. **Top bar**: project, undo, black or white, remote, MIDI, highlight, master, blackout
2. **Fixtures**: everything on the stage, in groups
3. **Sources**: effects, your shaders, images, video, capture, streams
4. **Tools**: select, line, matrix, circle, dot, path, pen, zones
5. **Stage**: your layout, playing live
6. **Dock**: Scenes · Patch, with Discovery, the DMX monitor and Output settings
7. **Right panel**: Inspector · Content · Settings
8. **Status bar**: frame rate, pixels, universes, what goes out



TIP

Right-click anything for its options. H hides the right panel, F fits the stage.

03

A NEW PROJECT

START FROM AN EMPTY CANVAS

1

Project ▾ → New empty project. It starts with one zone playing Rainbow, so every pixel you draw lights up.

2

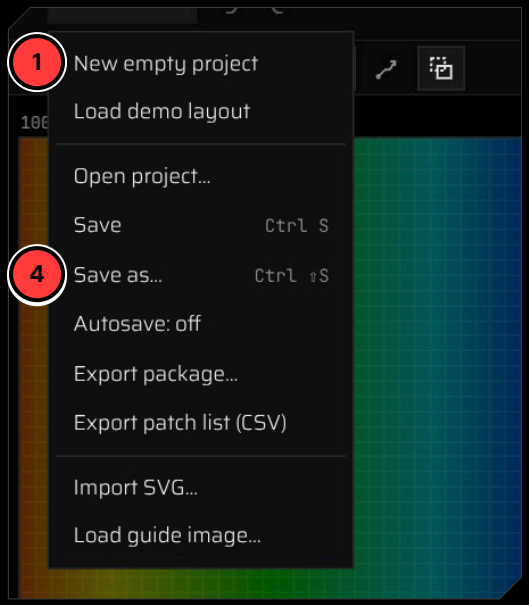
Name it: click the name in the top bar.

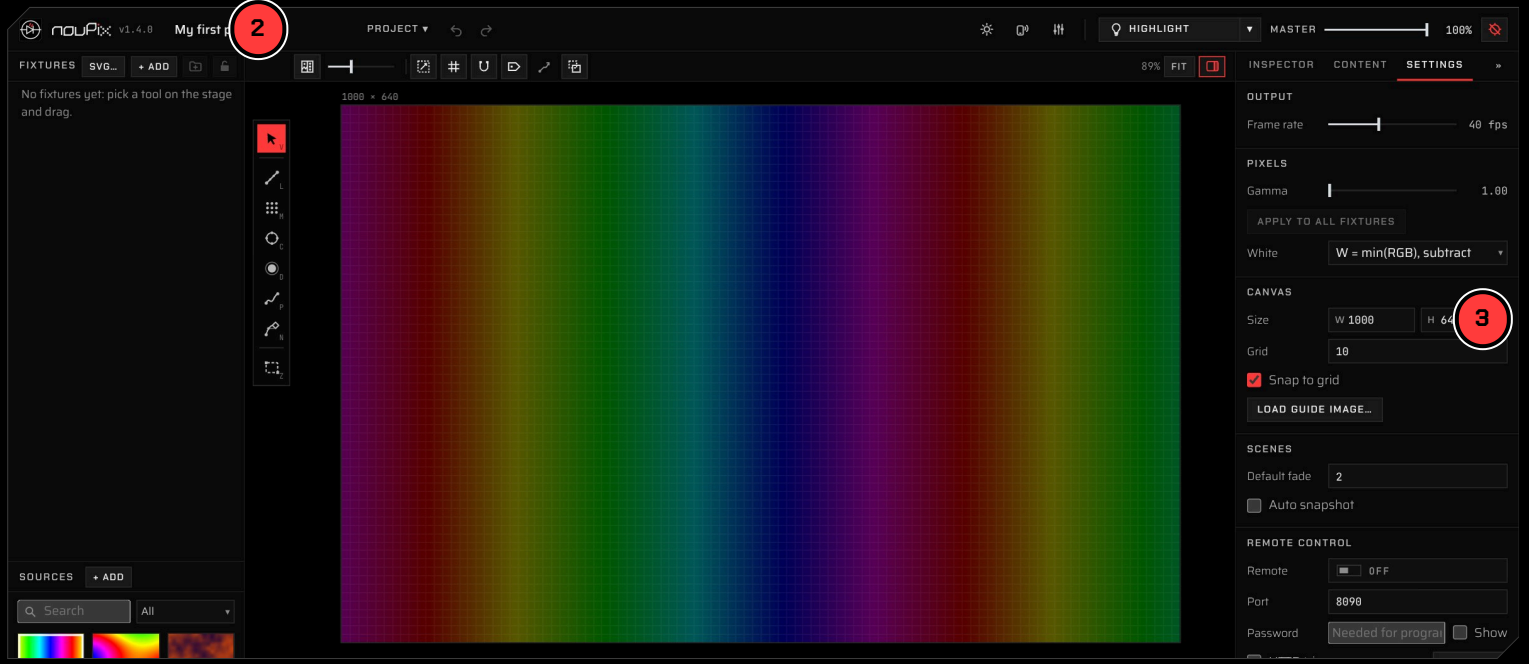
3

Settings → Canvas: the size of your canvas, in any unit you like (for example 1000 × 640). The frame rate is under Output.

4

Project ▾ → Save as... writes a .noupix file. Turn on Autosave and every change is saved to it.





TIP Ctrl S saves, Ctrl Z undoes. Projects open with Project ▾ → Open project...; the examples on GitHub are ready to open.

04

DRAW FIXTURES

DRAW WHAT YOU BUILT

1

Pick a tool: `L` line or strip, `M` matrix, `C` circle or ring, `D` dot, `P` path, `N` pen for curves.

2

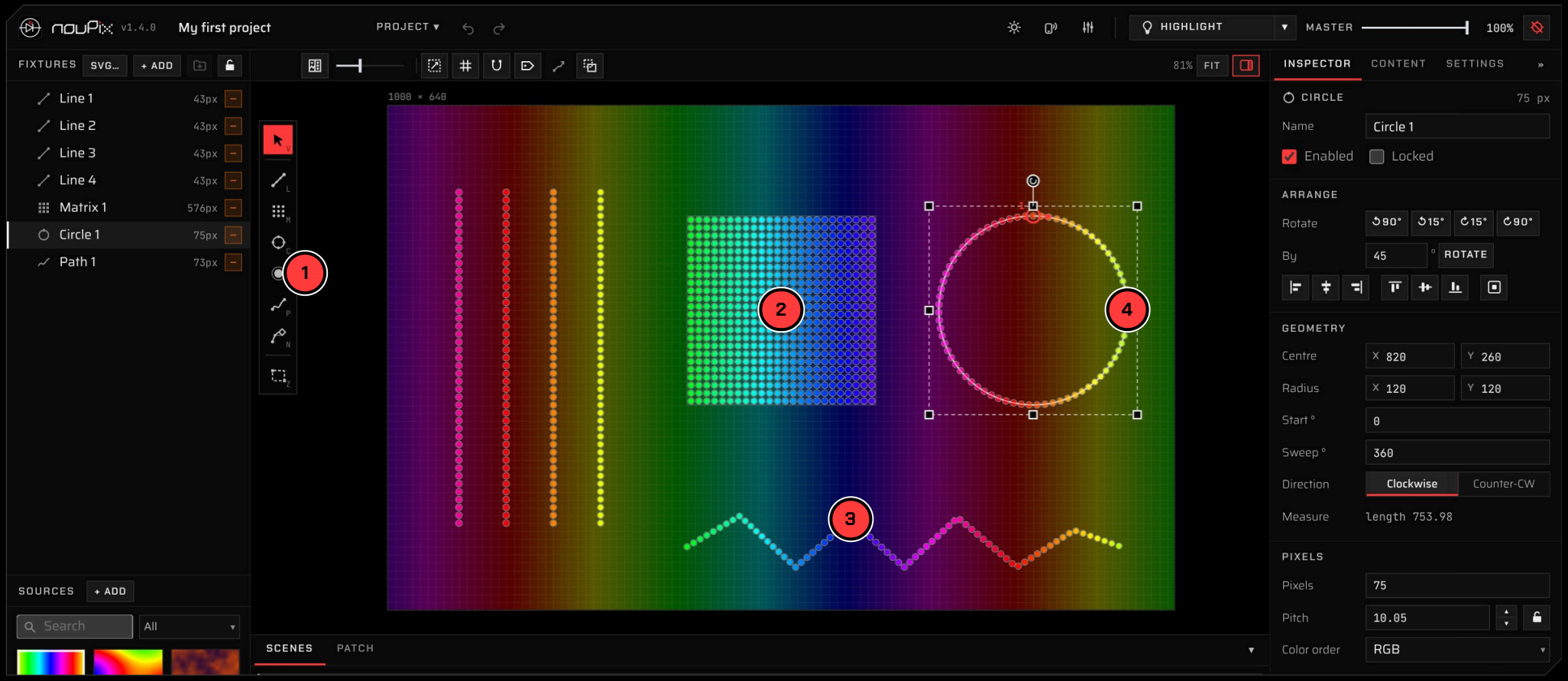
Drag on the stage: a line from end to end, a matrix as a rectangle, a circle from its centre outwards.

3

Path: click the points, then `Enter`. Pen: click for corners, drag for curves.

4

`V` selects and moves. `Shift` snaps to 45°, `Alt` turns snapping off. A click with a tool drops a default fixture.



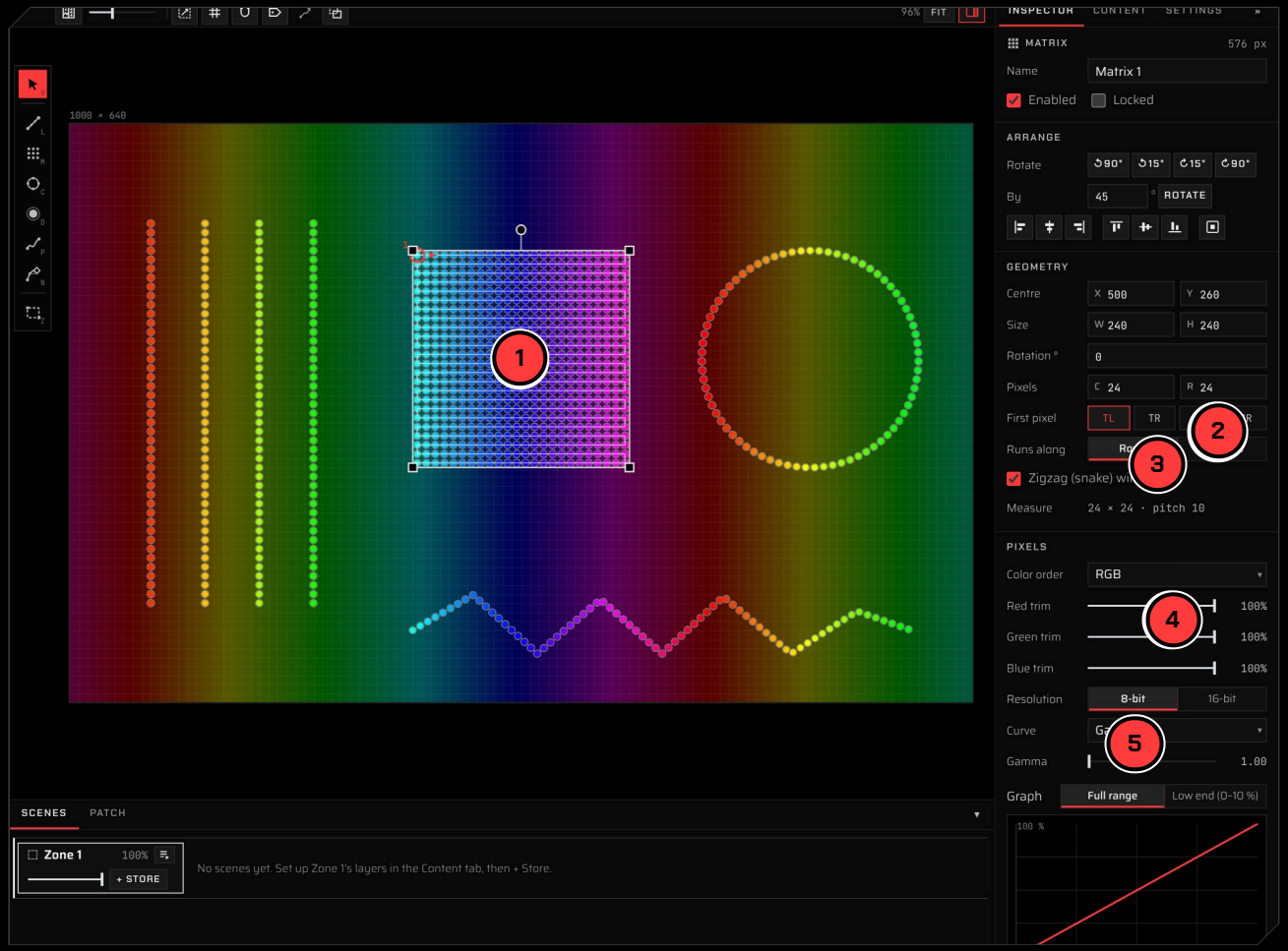
TIP Each new fixture appears in the Fixtures list on the left, with its pixel count. A Dot is one pixel as big as its light (a PAR, a wall washer): set its Size in the Inspector.

05

SET THEM UP

PIXELS, WIRING, COLOUR

1. **Select a fixture**: its settings open in the Inspector.
2. **Pixels**: the count, or columns × rows for a matrix.
3. **First pixel and wiring**: which corner, rows or columns, zigzag.
4. **Colour order** (RGB, GRB, RGBW…) and trims per colour.
5. **Resolution** 8 or 16-bit, and the dimming curve with its graph.
6. **Pitch lock** on a strip or ring: its length sets the pixels, like a strip cut to length.



LINE 1 · PITCH LOCKED



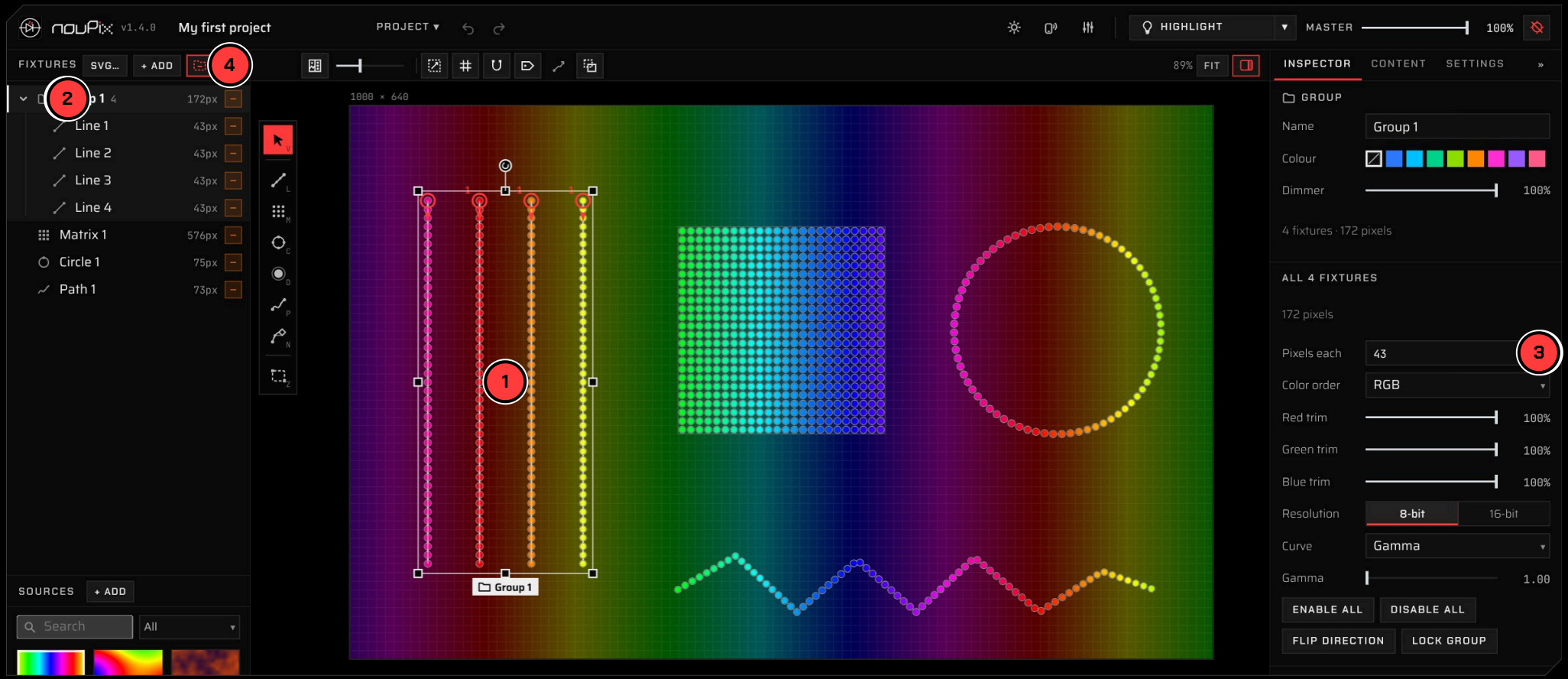
TIP Select several fixtures to set their pixels, colour order or gamma all at once.

06

ARRANGE AND GROUP

KEEP THE LAYOUT TIDY

1. Select several: `Shift` -click, or drag a box around them.
2. Group them with `Ctrl` `G` : the list folds them under one name, with its own dimmer.
3. Rotate, align and set pixels for all of them at once.
4. Lock with `Ctrl` `L` so nothing moves by accident.
5. Copy, paste and duplicate: `Ctrl` `C` `Ctrl` `V` `Ctrl` `D` .



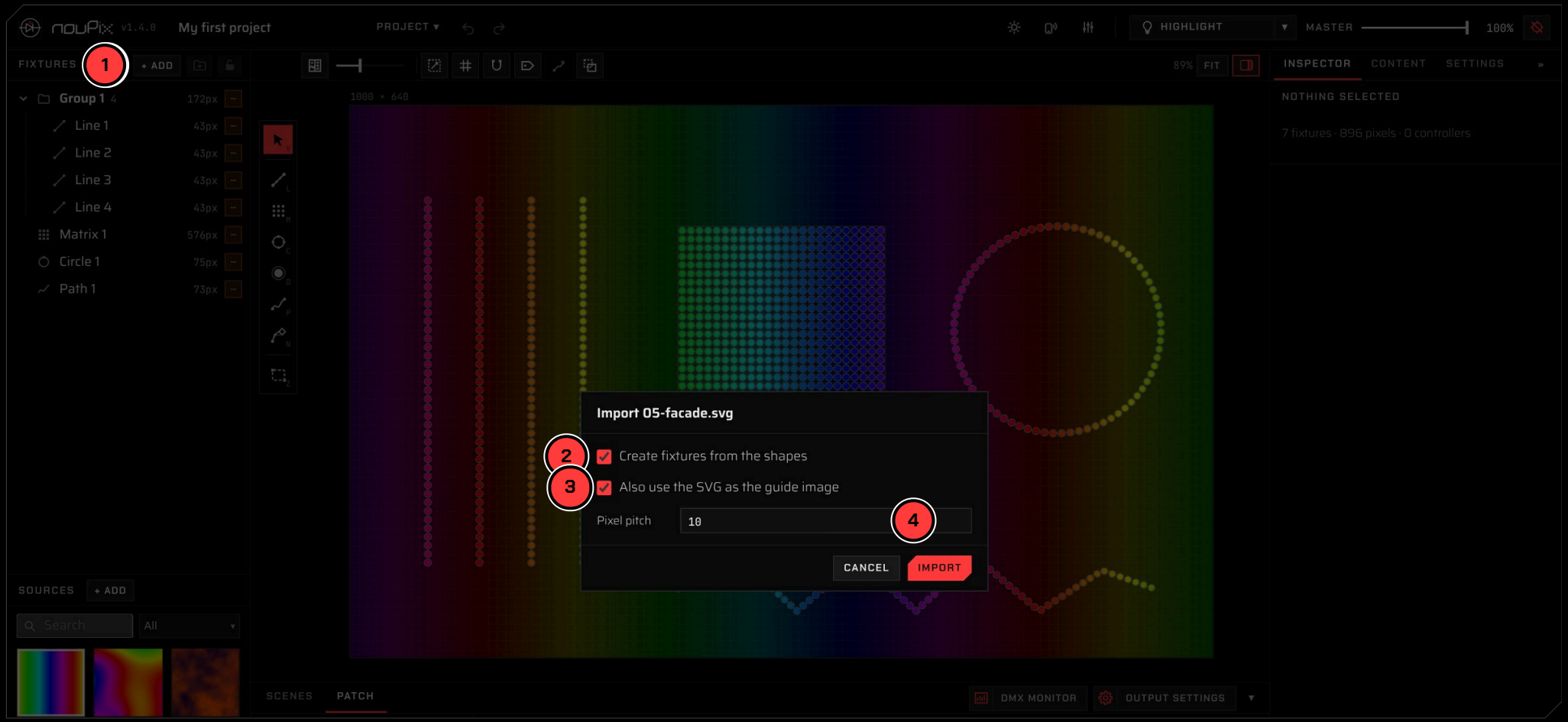
TIP Double-click a fixture inside a group to pick just that one. `R` turns the selection 90°.

07

IMPORT A DRAWING

START FROM YOUR SVG

1. SVG… at the top of the Fixtures list, or Project ▾ → Import SVG…
2. Create fixtures: every circle, arc, line and path of the drawing becomes one.
3. Use it as the guide image, under the fixtures, to check them against the building.
4. Pixel pitch: the spacing of the LEDs; it sets each fixture's pixel count.



TIP Export the façade from your drawing or 3D software as SVG, one stroke per LED strip.

08

FIND CONTROLLERS

DISCOVER THE DEVICES

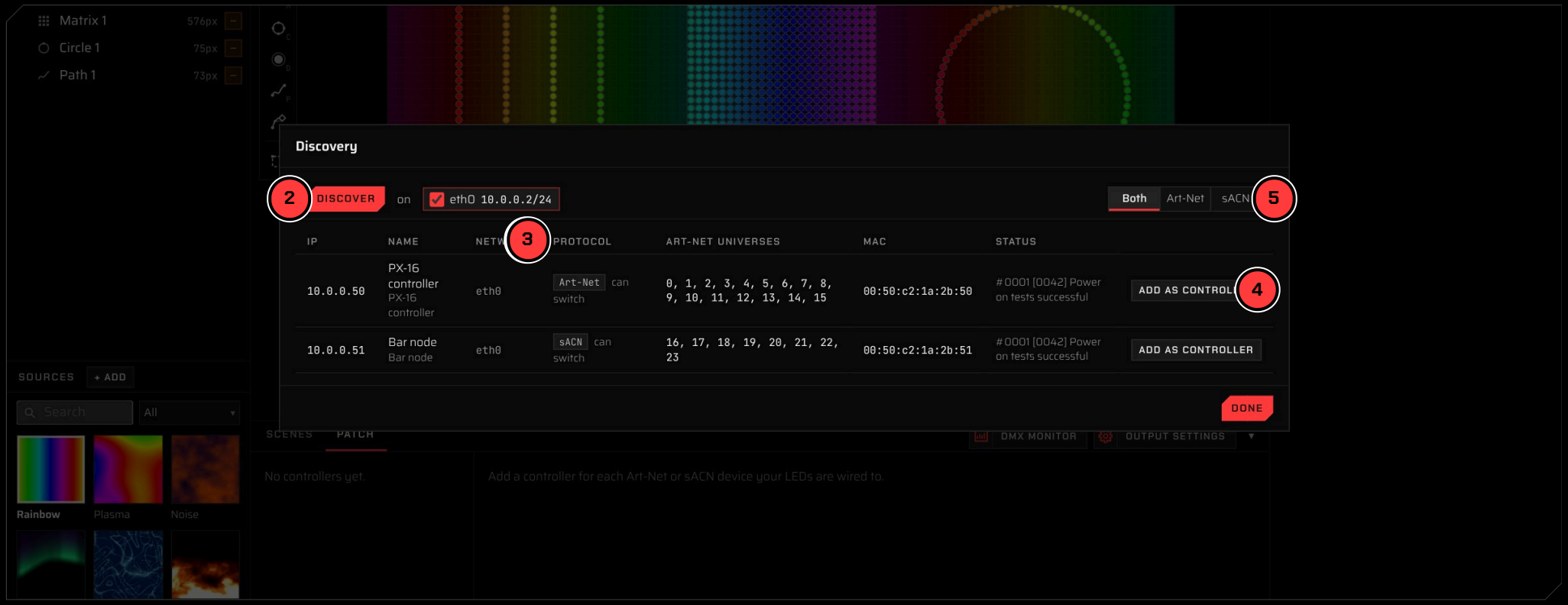
1. Connect the controllers and this computer to the **same network**, with addresses in the same range (for example 10.0.0.x, or 2.x.x.x).

2. Patch tab → **Discovery…** → **Discover**: ArtPoll goes out on the networks ticked next to it.

3. Every device that answers is listed with its **IP**, **name**, **protocol** (Art-Net or sACN) and universes.

4. **Add as controller**: it goes to the Patch list on its protocol, ready to patch.

5. **Both** · **Art-Net** · **sACN** shows only the devices for one protocol.



TIP Not found? Allow nouPix through the firewall on private networks, and give this computer an address in the device's range. **+ Controller** adds one by hand.

09

PATCH

FIXTURES ONTO OUTPUTS

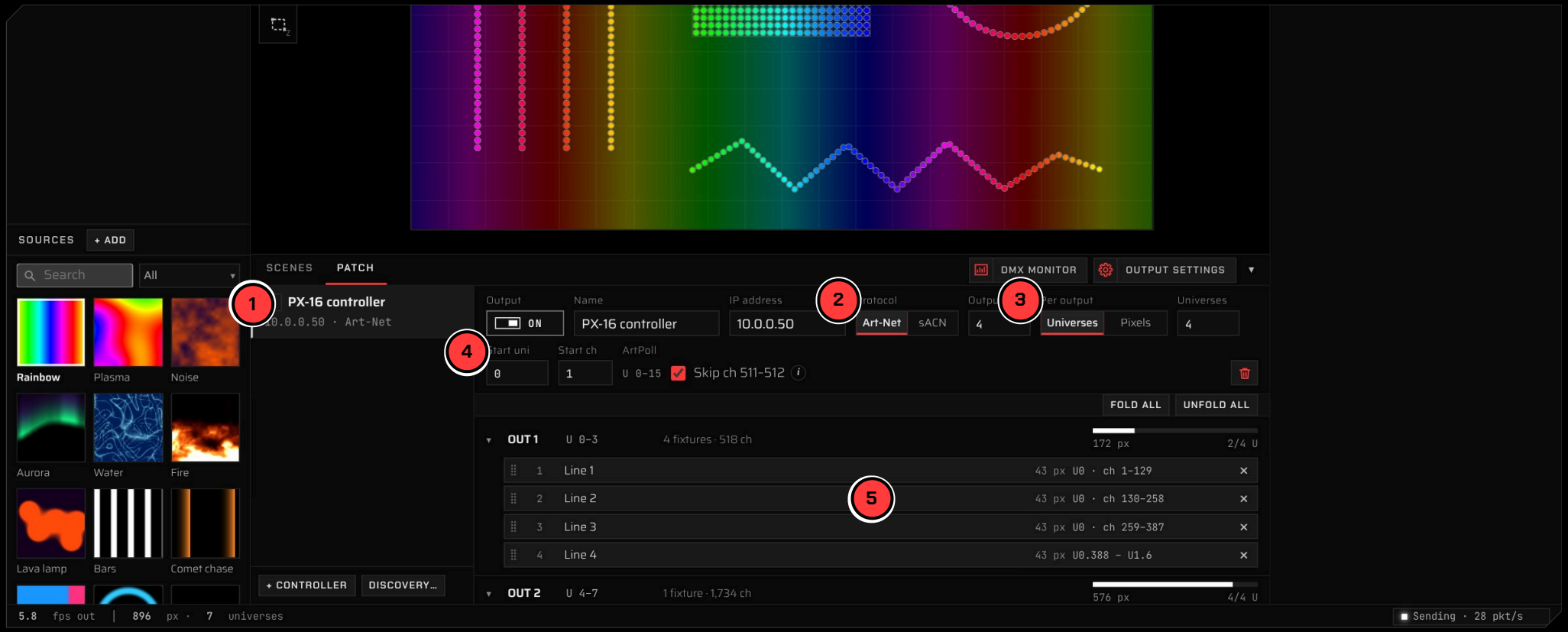
1 The **controller list**: name, IP and protocol. `Ctrl` or `Shift` selects several, `Del` removes them.

2 **Protocol** per controller: Art-Net and sACN go out together.

3 **Outputs** by **Universes**, or by **Pixels**: give the pixels per output and the pixel type, nouPix works out the universes.

4 **Start uni** and **Start ch**: where the controller begins.

5 **Drag fixtures** from the list onto an output. The order on an output sets the addresses.



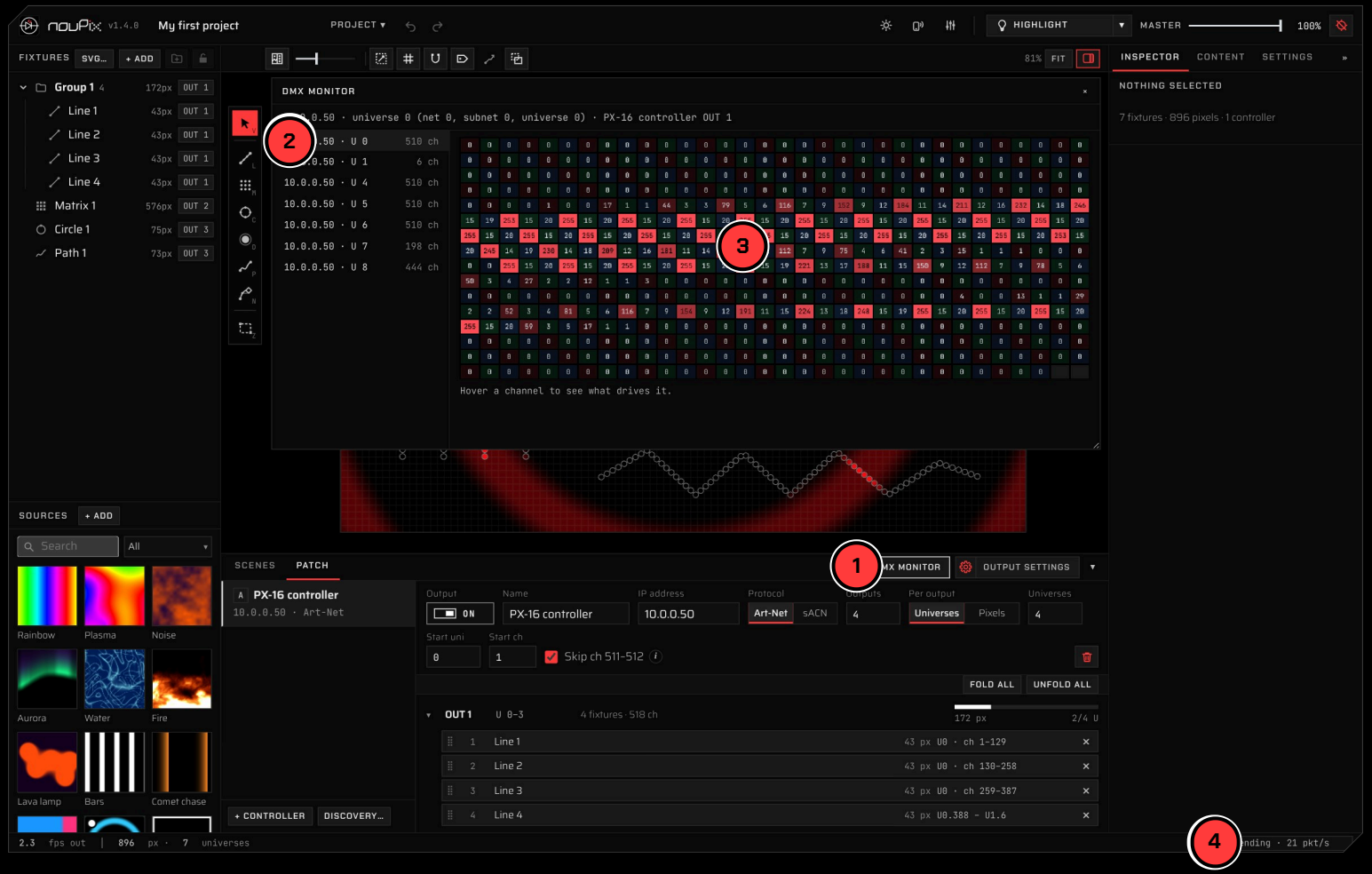
TIP Right-click a fixture → **Patch to…** The badge next to each fixture in the list shows its output.

10

DMX MONITOR

CHECK WHAT GOES OUT

1. DMX monitor at the right of the Patch tab: a window over the app, so you keep working.
2. Pick a universe: its 512 channels, live.
3. Point at a channel to see which fixture and pixel drive it.
4. The status bar shows how many packets go out each second.



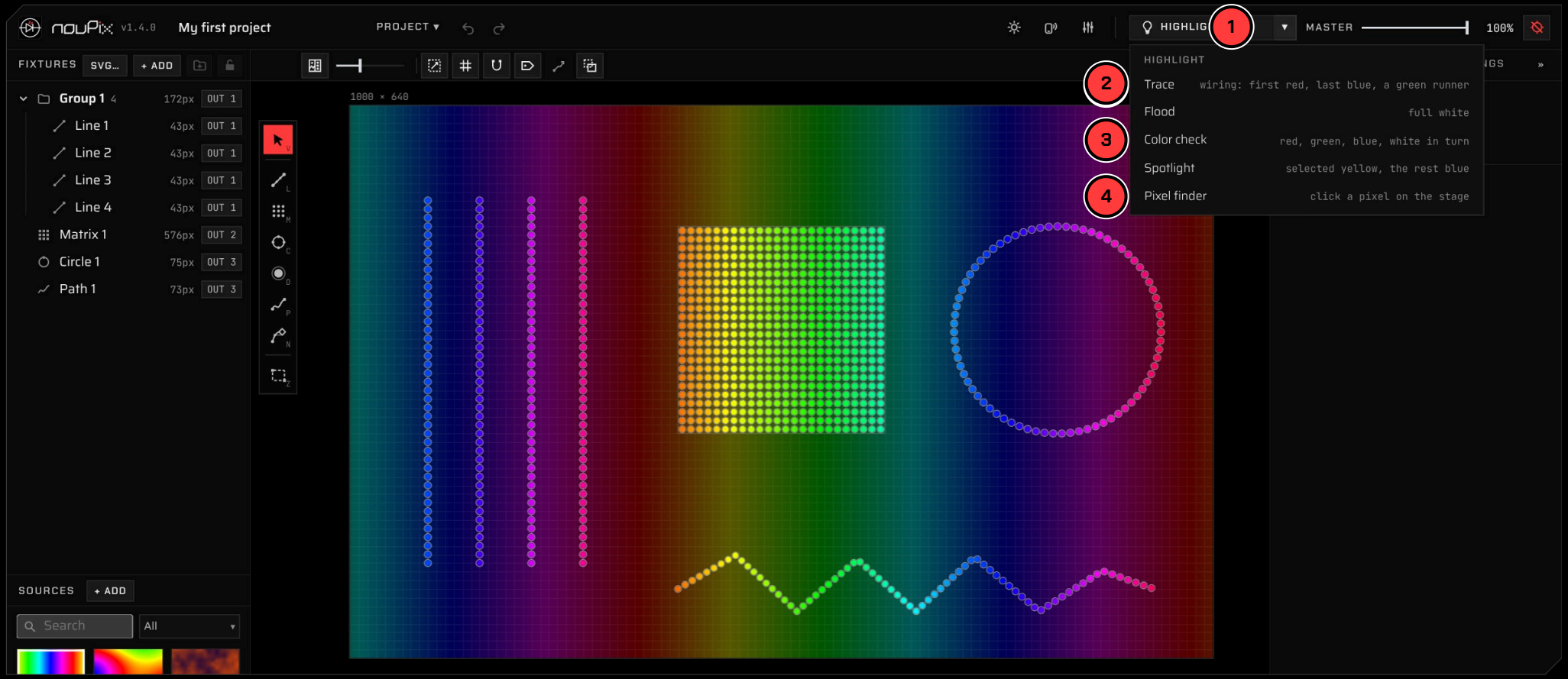
TIP Patched fixtures light the monitor even with no controller on the network: check the patch at your desk, before the site. Drag the window by its title.

11

HIGHLIGHT ON SITE

CHECK EVERY FIXTURE

1. **Highlight** in the top bar, or `I`: the content stops and test patterns light the fixtures.
2. **Trace**: first pixel red, last blue, a green runner along the wiring.
3. **Flood**, **Color check**, **Spotlight** (the selected ones yellow, the rest blue).
4. **Pixel finder**: click a pixel on the stage to light only that one.



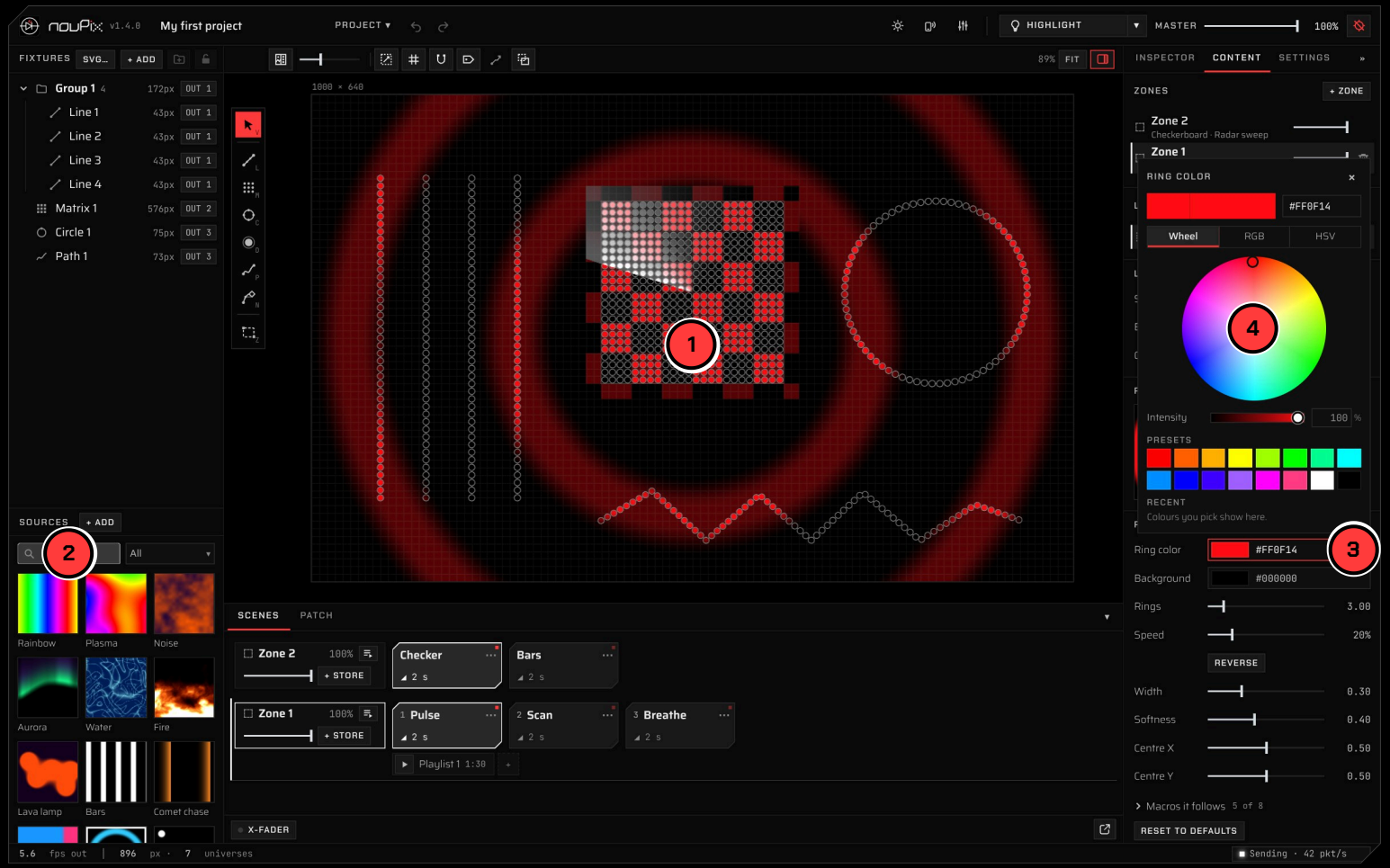
TIP Trace shows at once where a strip starts and whether it runs the right way.

12

PLAY CONTENT

CLICK A SOURCE, IT PLAYS

1. **Click a source**: it plays on the selected zone, and the stage shows it on the pixels.
2. **Search**, or show one use: Architectural, Ambient, Show, Utility.
3. Its **controls** in the Content tab. Double-click a value to type it.
4. **Colours**: wheel, RGB, HSV, presets and recent colours.



TIP `Shift` + scroll over a slider steps it; angles have buttons every 45°.

13

ZONES AND LAYERS

DIFFERENT CONTENT, DIFFERENT PARTS

1

Z Zones tool: drag a zone around part of the stage.

2

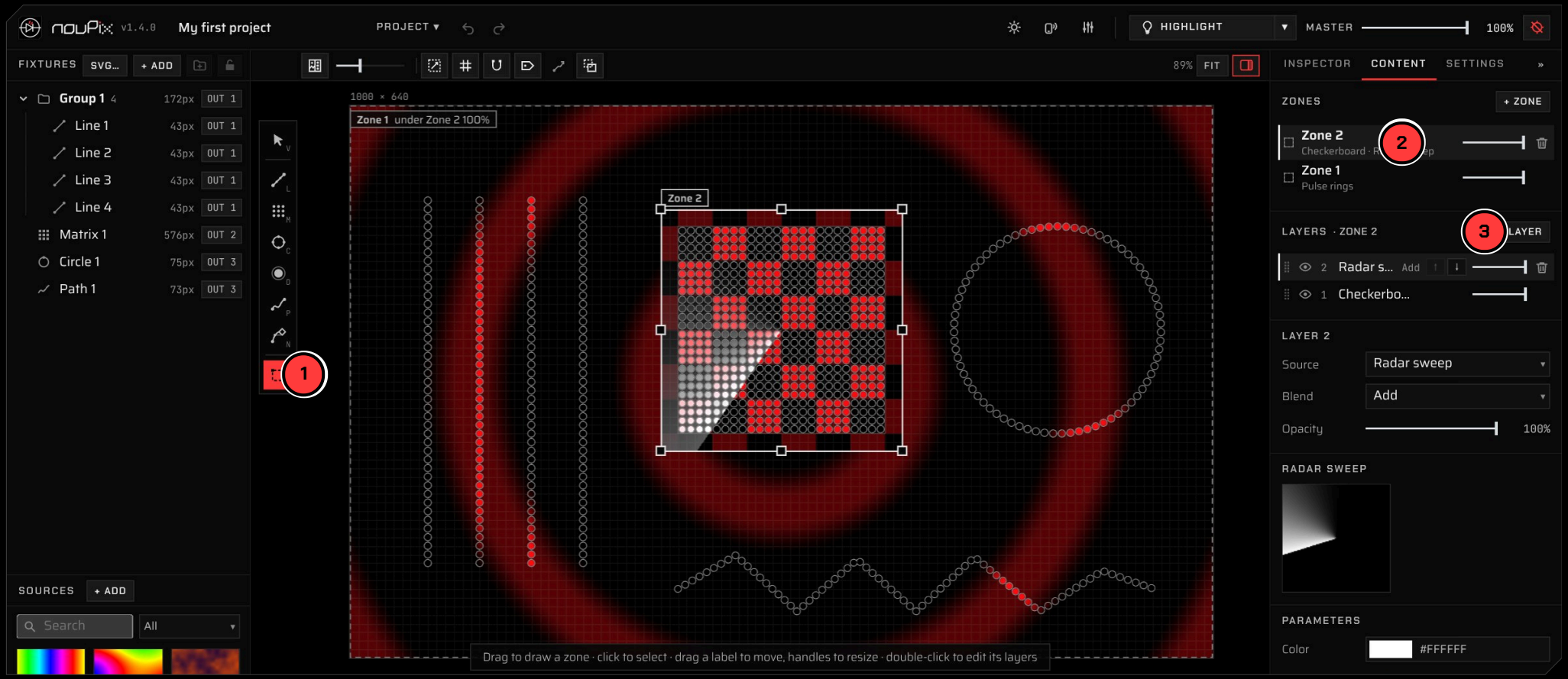
Zones list: the upper zone covers the lower. Drag to reorder.

3

+ Layer stacks another source in the zone.

4

Blend (Add, Screen, Multiply, Lighten…) and opacity per layer.



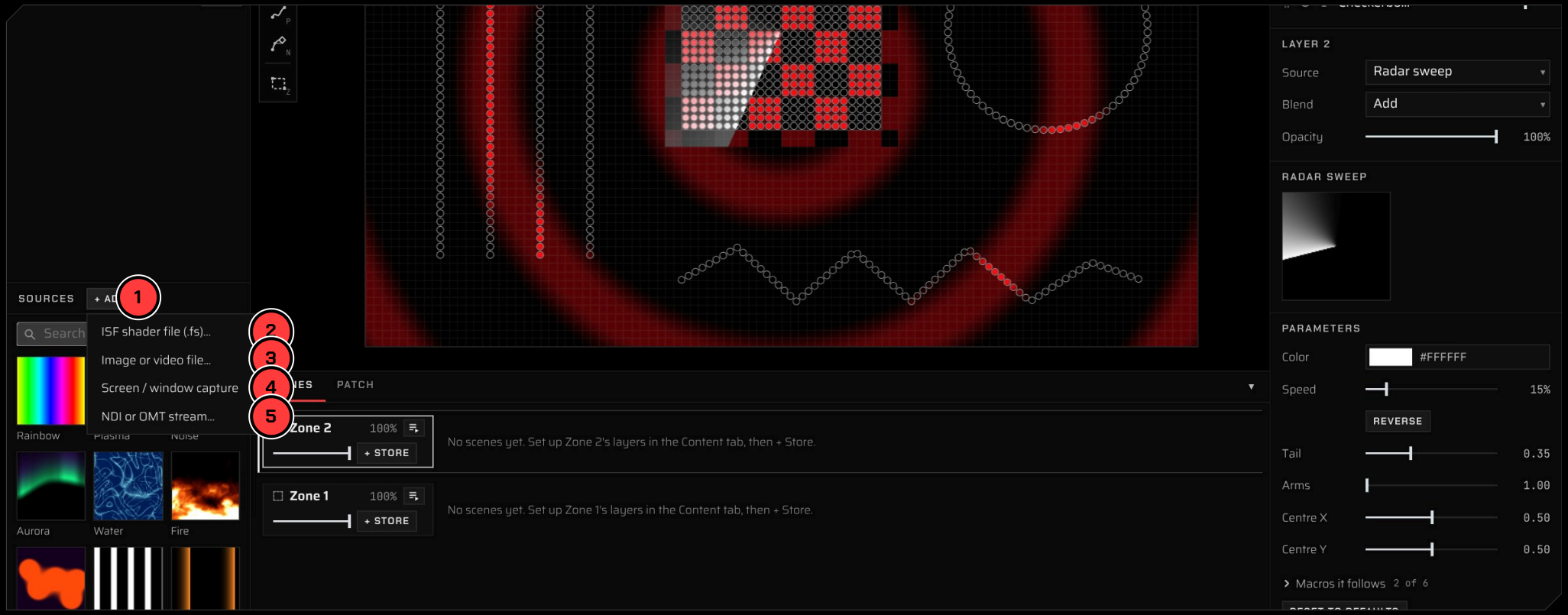
TIP Here Zone 2 plays a red checkerboard with a white radar sweep added on top, over the matrix only.

14

YOUR OWN CONTENT

SHADERS, VIDEO, CAPTURE, STREAMS

1. + Add above the sources.
2. ISF shader file (.fs): your own generators, with their controls.
3. Image or video: play, pause, trim, loop or once, with or without sound.
4. Screen / window capture: for example the output of other VJ software.
5. NDI or OMT stream: from Resolume, TouchDesigner, OBS or another computer.



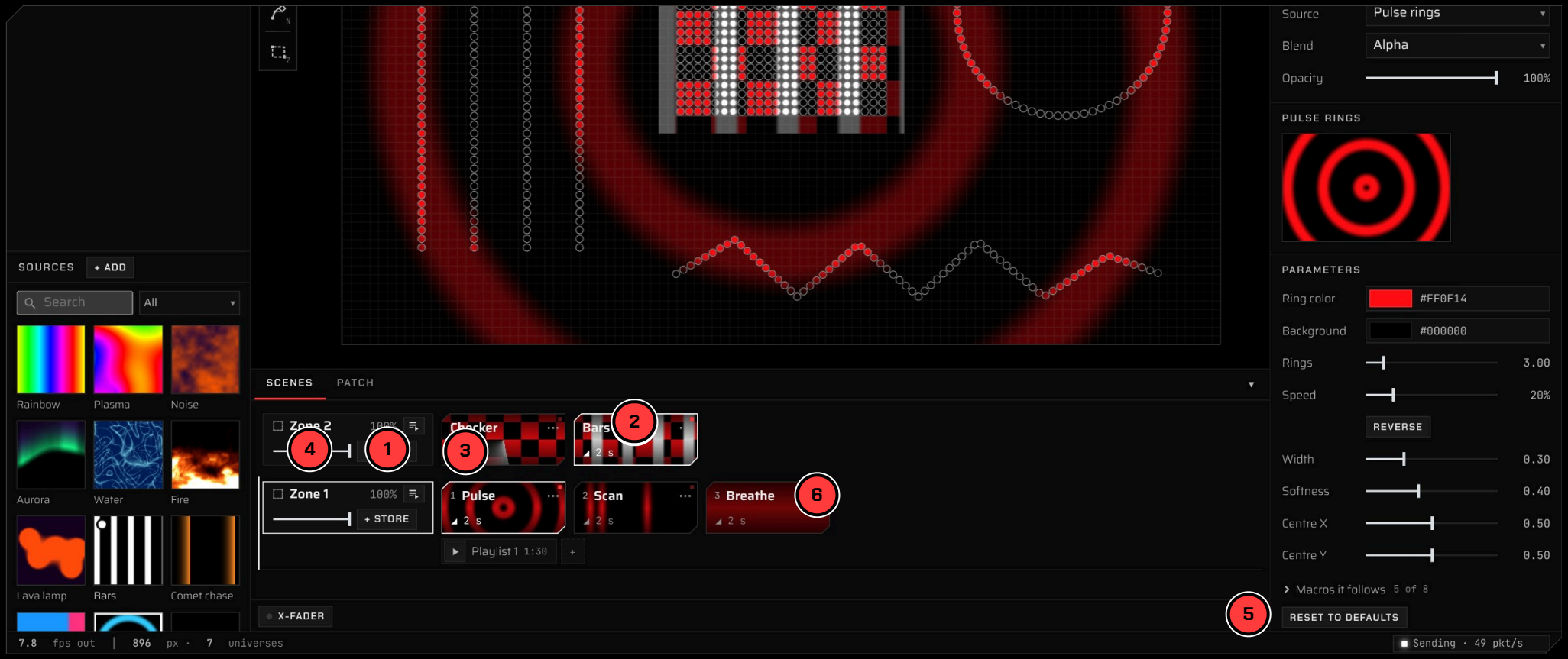
TIP Images and videos are copied into nouPix's media library, so projects open with them. NDI needs NDI Tools installed, OMT its library (see the user guide).

15

SCENES

STORE A LOOK, FADE BACK TO IT

1. **+ Store** keeps the zone's layers as a scene.
2. Click a scene to fade to it; `Shift` - click cuts.
3. Drag the fade time on its button to change it.
4. Level of each zone, live.
5. Open the **Scenes window** for a second screen.
6. ⋯ → Take **snapshot**: shown on its button.



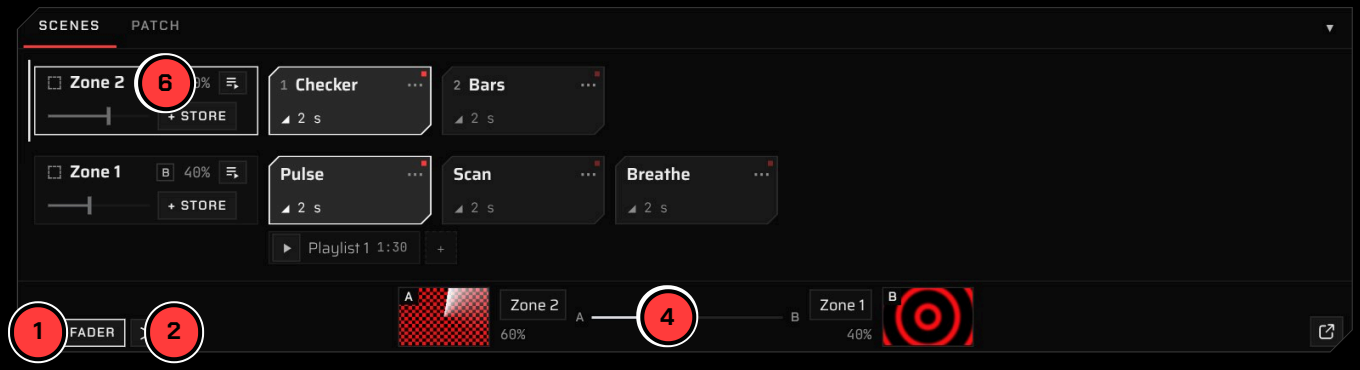
TIP Keys `1`-`9` play the scenes of the selected zone. Settings → Scenes → **Auto snapshot** takes one on every + Store.

16

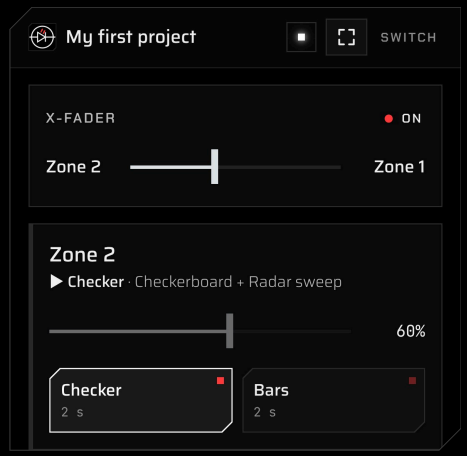
X-FADER

ONE FADER BETWEEN TWO ZONES

1. **X-Fader**, bottom left of the Scenes panel, turns it on. **Make both full size** when asked: A and B then cover every light.
2. **Mode**: Mix, Dip through black, or Cut at the middle.
3. **Live previews** of A and B, even the one faded out.
4. **The fader**: all A on the left, all B on the right.
5. **Choose the zones** for A and B from their lists.
6. **A and B tags**: while it is on, only the X-Fader sets their levels.







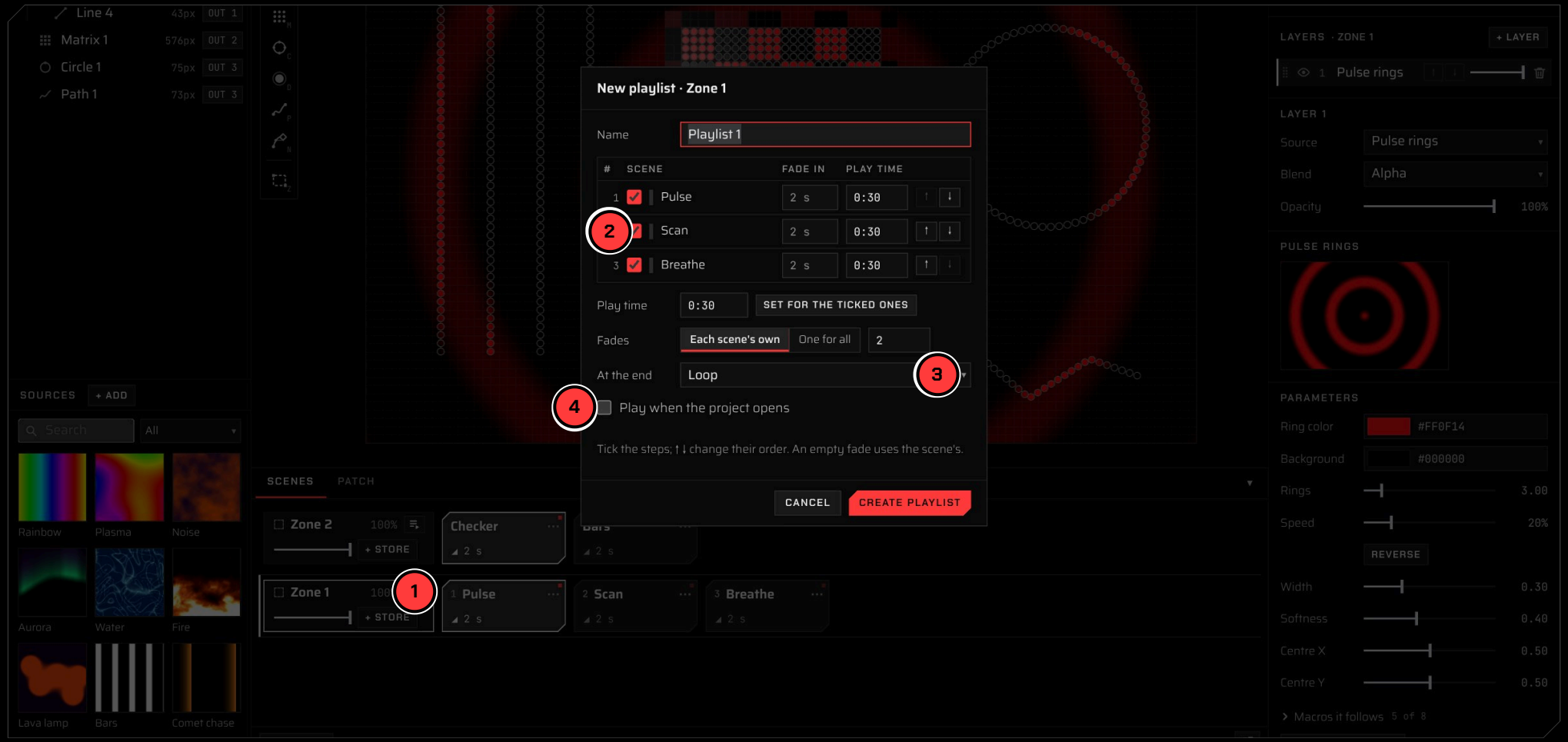
TIP Learn a MIDI fader onto it like any slider. Settings → Remote control → **X-Fader on the Remote page** puts it on the phones too.

17

PLAYLISTS

A SHOW THAT RUNS BY ITSELF

1. The playlist button next to + Store opens New playlist.
2. Pick the scenes, their play times and fades.
3. Play, pause, next. In a loop, once, or handing over to another playlist.
4. Tick Play when the project opens: the show starts by itself.



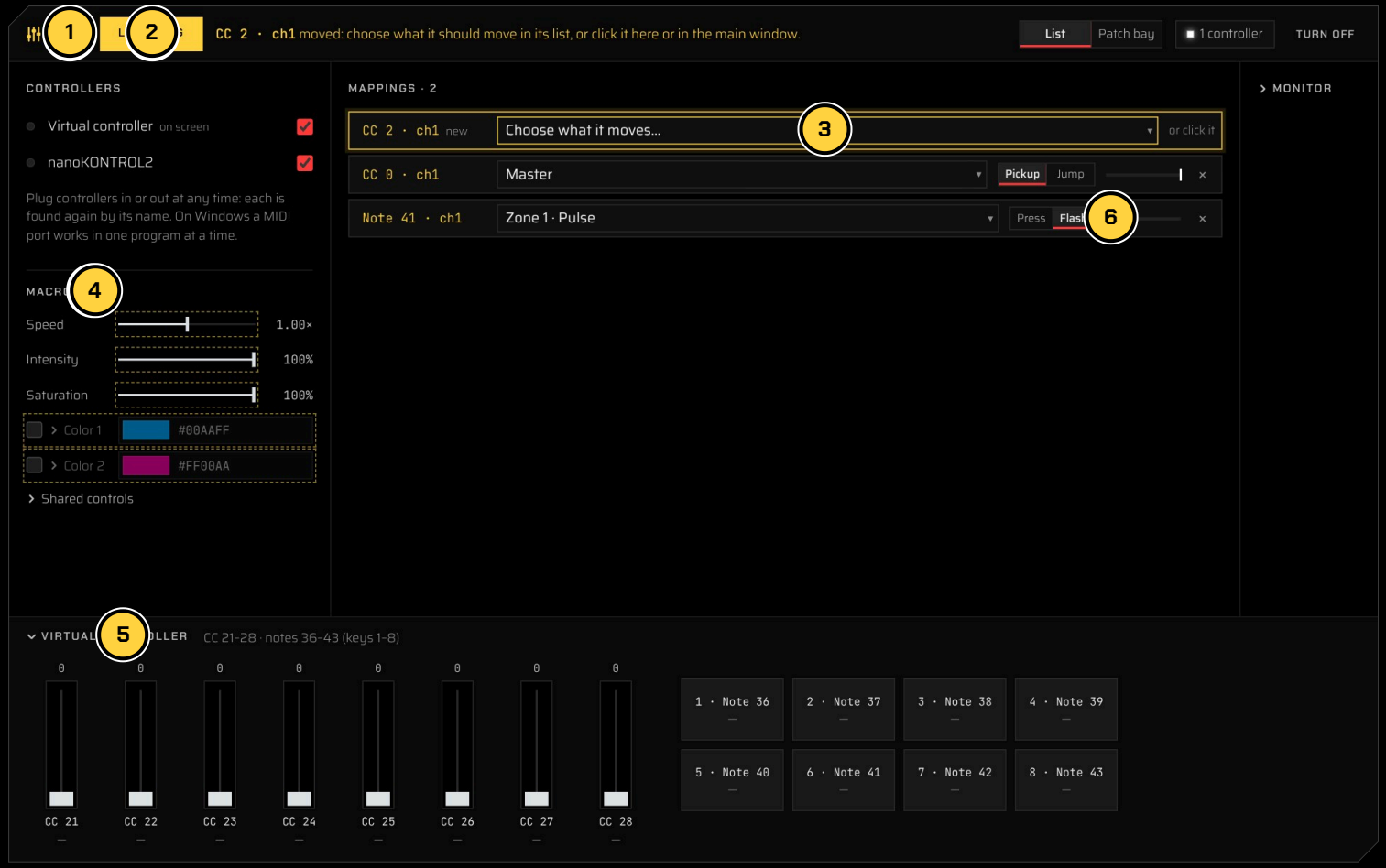
TIP A zone can have several playlists; the phone remote plays them too.

18

MIDI

LEARN YOUR CONTROLLER

1. MIDI in the top bar opens the MIDI window, with your controllers.
2. Learn, then move a fader, knob or pad.
3. Choose what it moves from the list, or click it in nouPix. Faders pick up where the slider is.
4. Macros: Speed, Intensity, Saturation and two colours, for every effect at once.
5. Virtual controller: faders and pads on screen, to try it without hardware.
6. Flash on a pad: a scene, a level or blackout only while held.



TIP Master, blackout, the X-Fader, group dimmers, zone levels, scenes, playlists, macros and any control of an effect can be learnt.

19

PATCH BAY

EVERY MAPPING AS A WIRE

1

Patch bay: the same mappings, drawn as yellow wires.

2

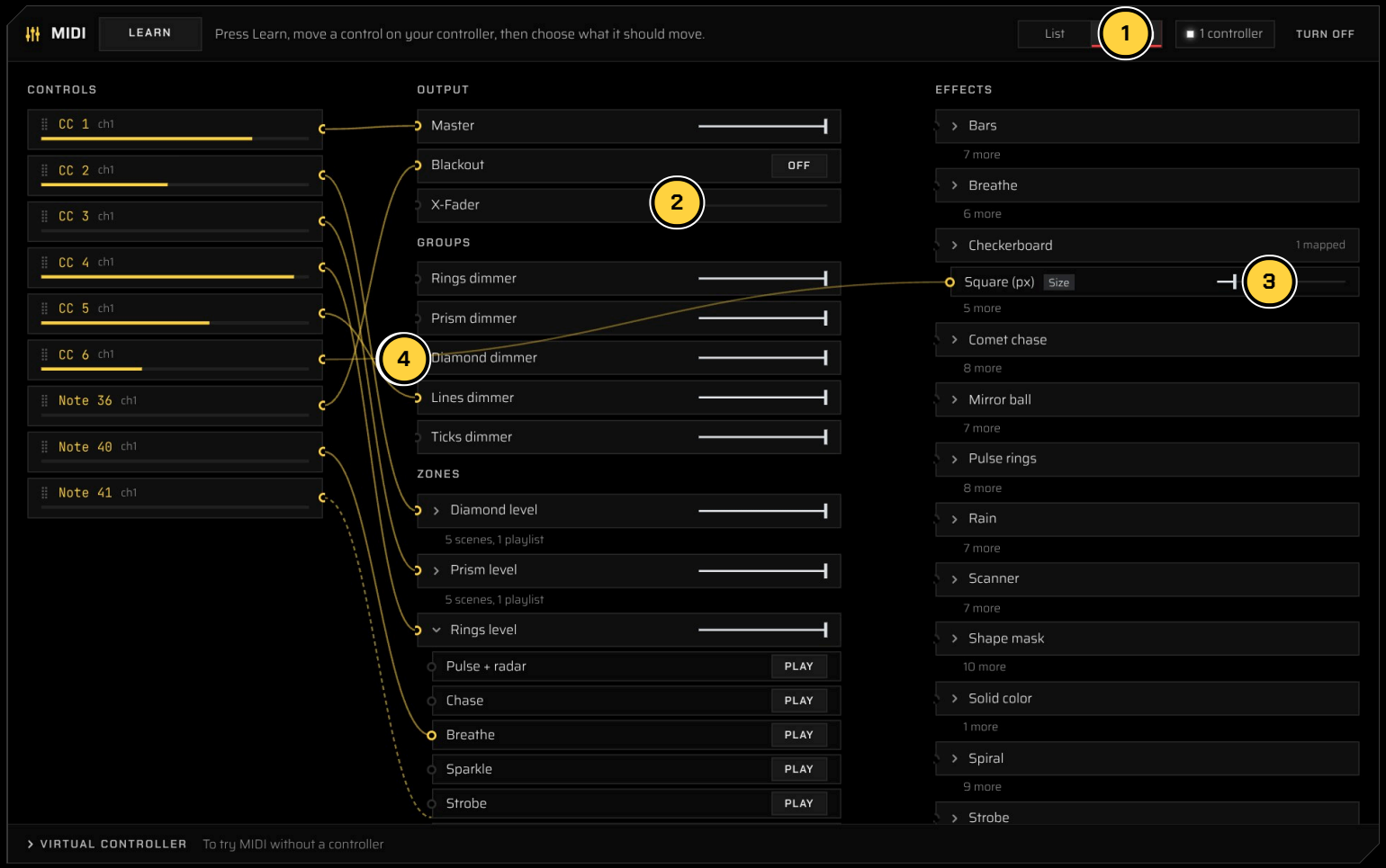
Controls on the left, what they drive in the middle: master, groups, zones, scenes, macros.

3

Effects on the right: wire any control of any effect.

4

Drag from a dot to another to wire them. Click a wire for its settings; Delete removes it.



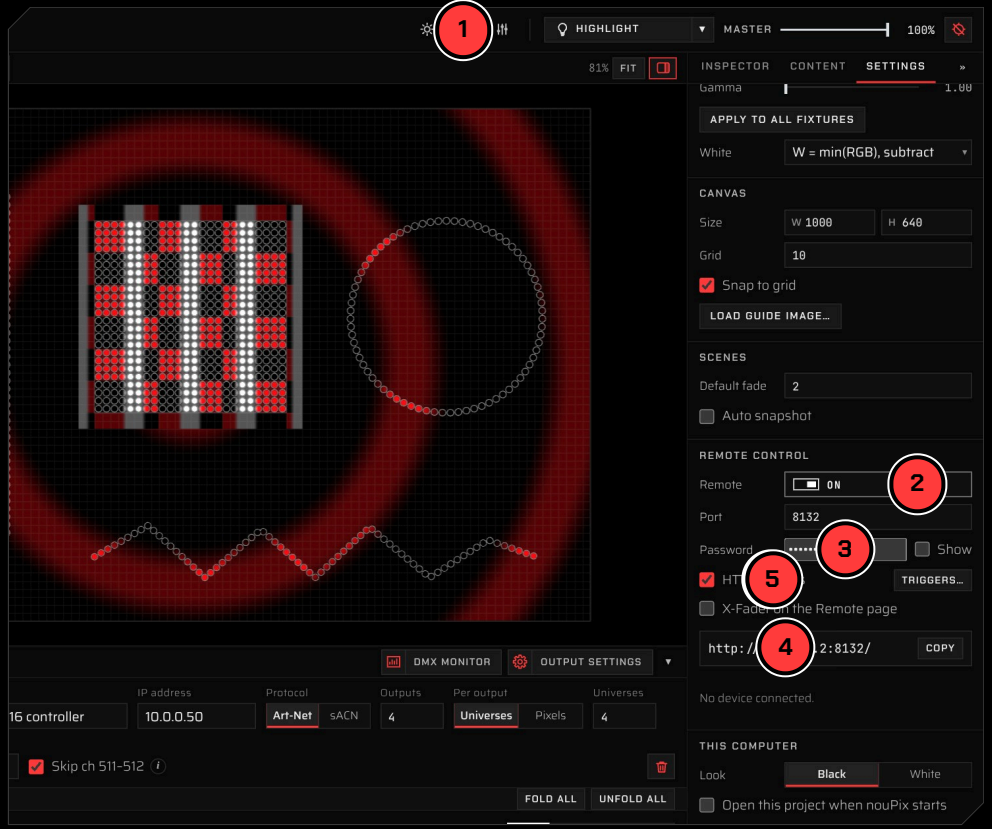
TIP Point at a macro to see, in aluminium wires, the effect controls that follow it.

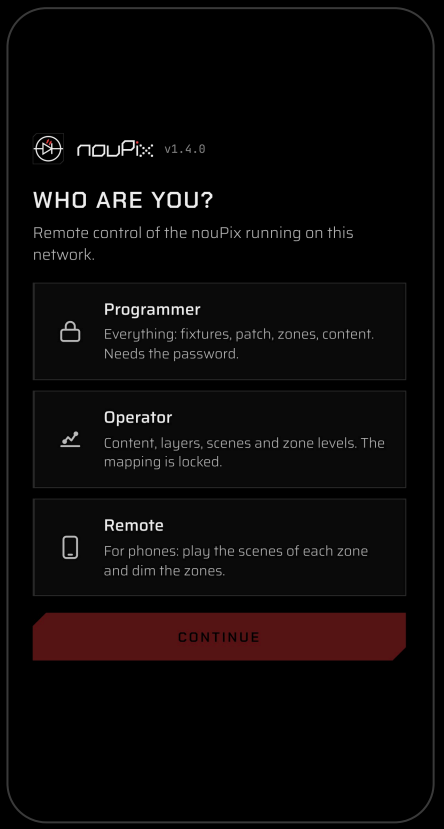
20

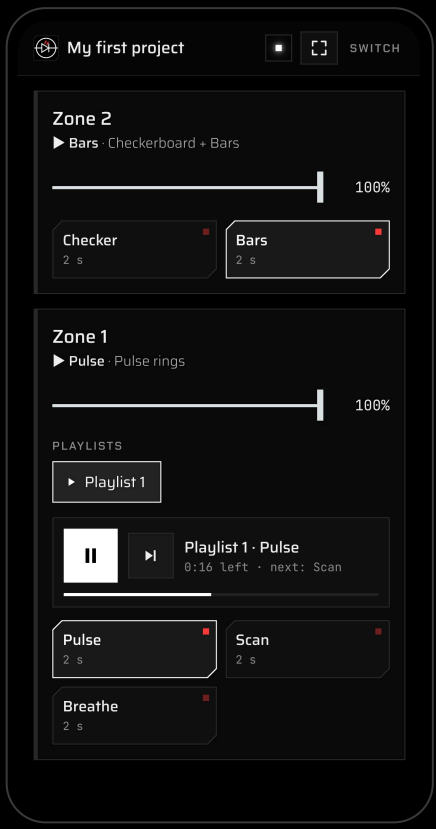
REMOTE CONTROL

HAND A PHONE TO THE END USER

1. **Remote** in the top bar → **Switch remote control on**, or Settings → Remote control.
2. **On**: a second server opens on your network.
3. Set a **password** for the Programmer role.
4. Open the **address** on a phone on the same network. **Remote** plays scenes and playlists, dims zones and, if you tick it, moves the X-Fader; **Operator** changes content too; **Programmer** everything.
5. **HTTP triggers**: show control, a Stream Deck or a building management system plays scenes and sets levels with a web address.







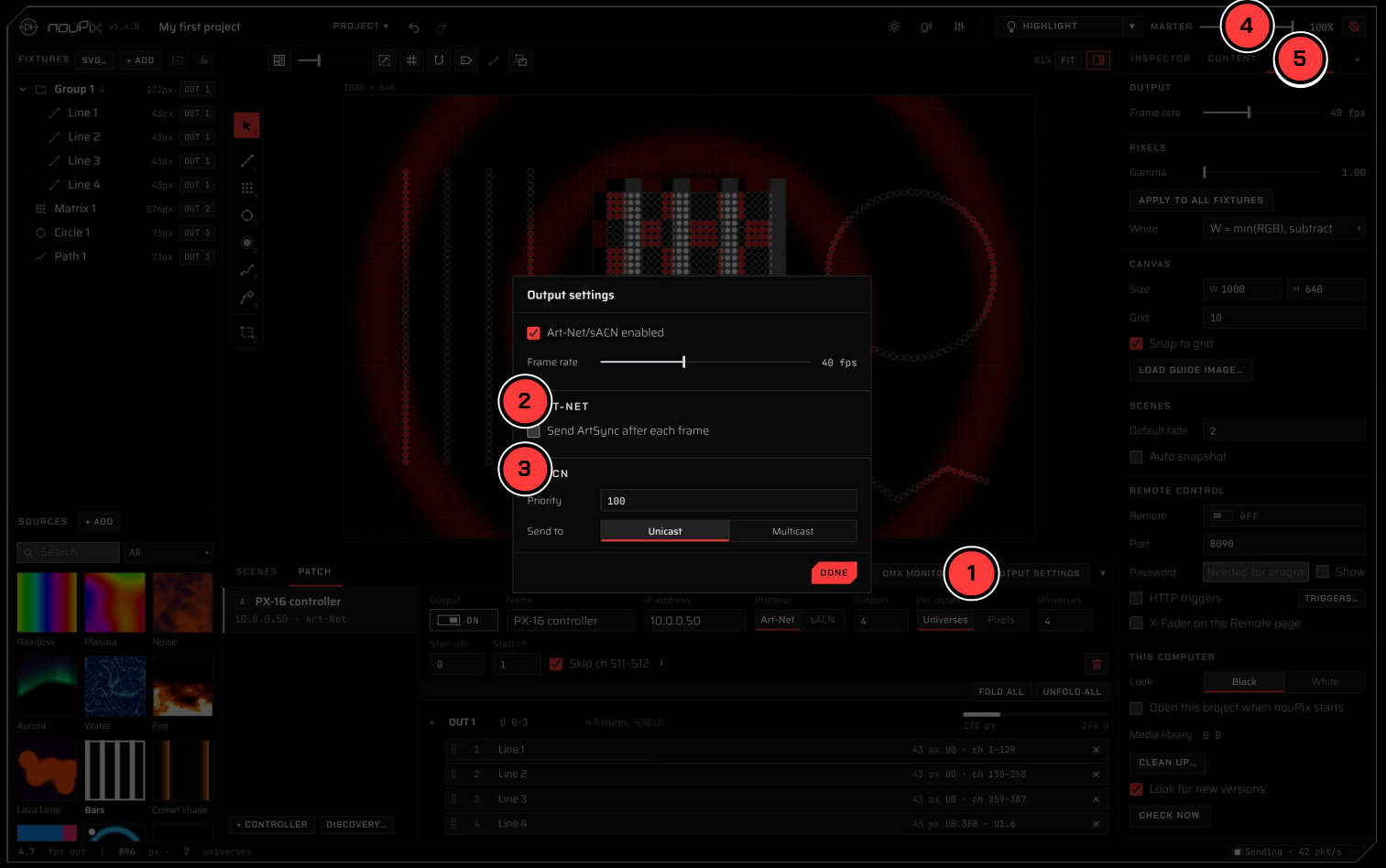
TIP With remote control on, **Remote** in the top bar → **HTTP triggers…** lists every address, ready to copy: scenes, zone levels, the master and blackout.

21

OUTPUT AND SETTINGS

MASTER, LOOK, ART-NET AND sACN

1. **Output settings** (the cog in the Patch tab): output on or off, the frame rate.
2. **Art-Net**: ArtSync after each frame, for nodes that support it.
3. **sACN**: priority, and unicast or multicast.
4. **Master** and **blackout** (B) in the top bar.
5. **Settings**: Pixels (gamma, RGBW white), Canvas, Scenes, and This computer: look, startup project, updates.



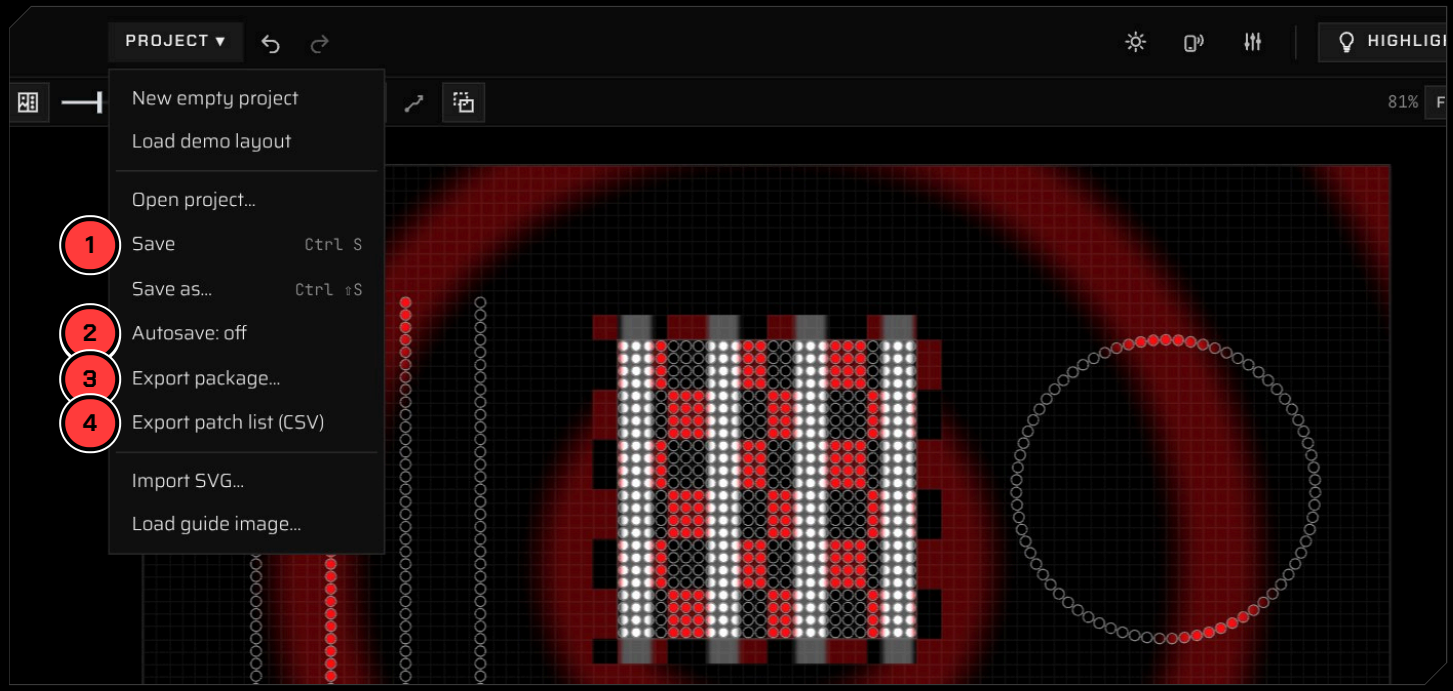
TIP Untick **Art-Net/sACN enabled** to stop the output, for example while a lighting console drives the lights. The light / dark button in the top bar switches the look.

22

SAVE AND HAND OVER

KEEP IT, SHARE IT

1. Save (`Ctrl` `S`) writes over the .noupix file; Save as... makes a new one.
2. Autosave writes every change to the file.
3. Export package...: one .noupixpkg file with the project and its images and videos, for another computer.
4. Export patch list (CSV): universes and addresses for the install crew.



TIP Settings → This computer → Open this project when nouPix starts, and a scene or playlist can play when it opens: a show computer runs by itself.

SHORTCUTS

KEYS TO REMEMBER

V	Select	L	Line	M	Matrix
C	Circle	P	Path	N	Pen
Z	Zones	Ctrl C	Copy	Ctrl X	Cut
Ctrl V	Paste	Ctrl D	Duplicate	Del	Delete
Ctrl G	Group	Ctrl Shift G	Ungroup	R	Turn 90°
Ctrl L	Lock	Ctrl Z	Undo	Ctrl S	Save
1-9	Scenes of the zone	I	Highlight	B	Blackout
F	Fit the view	H	Hide the side panel	Space	Drag to pan
D	Dot	Ctrl Shift S	Save as	Ctrl Shift Z	Redo

On a Mac, Ctrl is ⌘.

DOWNLOAD
noupix.nouled.com

SOURCE AND MANUAL
github.com/nouled-studio/nouPix

BUGS AND IDEAS
github.com/nouled-studio/nouPix/issues

nouPix is free software under the GNU AGPL-3.0, made by NOULED.STUDIO and provided as it is, with no warranty. Art-Net™ designed by and copyright Artistic Licence Holdings Ltd.